FNU TUSHAR

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EXPERIENCE

Instructor and Grader, Advanced Database Systems, New York University — September 2026 - December 2026
New York, NY

- Led technical sessions for graduate students on AQuery, q/kdb+, and ReproZip, covering query execution, time-series data processing, and reproducible database experiments.
- Supported students with database-system assignments and projects involving transaction processing, performance tuning, indexing, and distributed systems.

Software Developer, Qdata, Sirius Holdings — April 2025 - May 2026
Abu Dhabi, UAE

- Implemented and tested custom MT5 trading algorithms; managed dozens of accounts with dashboards, automated multi-leg position management, and configurable execution and exit strategies.
- Built custom Terraform automation for 1,000+ concurrent AWS ECS workers and VPN connectivity between AWS and local servers, keeping restricted data on local servers.
- Optimized C++11 and C++17 SAT solvers and CNF generation; built SHA-256 and CORBA/OmniORB distributed systems.

Independent Software Consultant, Confidential Fintech Client — January 2026 - July 2026
Abu Dhabi, UAE

- Developed end-to-end marketplaces for a platform for the financialization and trading of real-world assets.
- Developed custom reinforcement-learning environments in Harbor, including task definitions, reward logic, and evaluation workflows for training and evaluating OpenAI and Anthropic models.

Backend and Data Engineer, Zator — December 2024 - March 2025

- Developed data streaming pipelines in Go and Rust to ingest JSON and Protobuf data from APIs and Solana RPC.
- Built Kafka streaming workflows with custom producers and consumers for real-time data filtering and processing.

Young Research Fellow, IIIT Hyderabad IHub — July 2024 - November 2024

- Built a real-time IoT system to collect sensor data with MQTT, Fastify, and SQLite and display machine learning results in interactive visualizations.
- Researched and implemented on-device ML inference in React Native using a custom ML Kit wrapper, react-native-fast-tflite for TensorFlow Lite, and onnx-runtime-react-native for ONNX.

EDUCATION

Master of Science in Computer Science, New York University — September 2026 - May 2028

Bachelor of Engineering in Information Science and Engineering
National Institute of Engineering, Mysore — January 2021 - August 2024

SKILLS

Programming languages: Python, Go, Rust, TypeScript, SQL, OCaml
Frameworks and libraries: PyTorch, React Native, Fastify, ML Kit, TensorFlow Lite, ONNX Runtime
Performance analysis: GPU roofline modeling, microbenchmarking, SIMD optimization
Databases: SQLite, Postgres, ClickHouse, q/kdb+, TigerBeetle
Tools and software: Kafka, Docker, Terraform, MT5, CORBA, OmniORB, Cargo
Cloud platforms: AWS: EC2, ECS, ECR, S3, IAM, VPC, VPN, Lambda, CloudWatch. Cloudflare: R2, Workers. Modal.

PROJECTS

GPU Napkin Math

- Built a Python roofline estimator and interactive Astro dashboard to model GPU compute, HBM traffic, host transfers, peer copies, kernel overhead, and memory capacity.
- Calibrated the estimator with PyTorch microbenchmarks on two Modal H100 SXM GPUs; recorded 717 TFLOP/s dense BF16 GEMM and 3.00 TB/s HBM copy bandwidth. Distinguished measured H100 results from vendor ceilings for B200/B300.

VLIW/SIMD Kernel Optimization

- Optimized Anthropic's simulated tree-traversal kernel from 147,734 to 2,910 cycles, a 50.8x speedup, using eight SIMD streams, register-resident state, software pipelining, and fused multiply-add instructions.
- Verified output correctness against the unchanged upstream frozen simulator and documented optimization steps and remaining gather-load bottlenecks.

Horizon: Reproducible Agent Experiments

- Built a TypeScript harness for agent-driven performance experiments with an independent verifier, immutable candidate snapshots, and durable SQLite/MongoDB experiment ledgers.
- Implemented checkpoint recovery, fenced controller leases, and scoped memory retrieval; added fault-injection tests for interrupted runs and checks that reject stale-cache and verifier-bypass candidates.

OCaml Ledger State Machine

- Reimplemented TigerBeetle ledger semantics in an experimental OCaml core with 128-bit accounting, two-phase transfers, linked-batch rollback, indexed queries, and pending-transfer expiry.
- Added property tests and equivalence scenarios using pinned upstream behavior as a reference; built workload benchmarks reporting throughput and allocation per operation.

Monkey Compiler

- Built an interpreter in Go with no external dependencies, supporting integers, booleans, functions, closures, conditionals, and lexical scoping.

Market Maker Simulator

- Built an asynchronous ETH/USDC market-making simulator in Rust. Ingested Binance WebSocket and Jupiter/CoW Protocol API prices concurrently and maintained the latest cross-venue market state through Tokio channels.
- Designed median-based quoting and probabilistic execution models with 2-second trading ticks, automatic WebSocket reconnection, dynamic fill probabilities, and real-time PnL and notional tracking.